

Exercise 5

Exercise 5.1 Singular case

The following dynamic optimization problem

$$\min_{u(\cdot)} \int_0^T (x - y)^2 dt \quad (1a)$$

$$\text{s.t. } \dot{x} = -ux, \quad x(0) = 0.1 \quad (1b)$$

$$u \in [u_{\min}, u_{\max}] \quad (1c)$$

is given w.r.t. a time-varying trajectory $y(t) = 0.5t$.

- Calculate the optimal trajectory of the control variable u . What is the degree of singularity?
- A work around for avoiding the singular case is to extend the cost function by a regularization term $\frac{c}{2}u^2$ with parameter c . What is the optimal solution for this case? Plot the trajectories for the different parameter values $c \in \{1 \cdot 10^{-4}, 5 \cdot 10^{-5}, 1 \cdot 10^{-5}, 5 \cdot 10^{-6}\}$ using the Matlab script E5_1_run_bvp and interpret the results. What result would you expect for $c \rightarrow 0$?

Exercise 5.2 Linear quadratic regulator (LQR)

Consider the nonlinear damped pendulum with system dynamics

$$\dot{\theta}(t) = \omega(t) \quad (2a)$$

$$\dot{\omega}(t) = -\frac{g}{l} \sin(\theta(t)) - \frac{b}{m l^2} \omega(t) + \frac{1}{m l^2} u(t), \quad (2b)$$

using the parameters $g = 10$, $l = 0.5$, $m = 2$ and $b = 0.5$. The system should be linearized around the operating point $\bar{x} = [\pi/2, 0]^T$ and $\bar{u} = 10$.

- Choose $x(t) = [\theta(t), \omega(t)]^T$ and write the system dynamics in state-space form. Then linearize it around $(\bar{x}, \bar{u})$ and state the linear model in the form

$$\Delta \dot{x}(t) = A \Delta x(t) + B \Delta u(t). \quad (3)$$

- Consider the infinite-horizon optimal control problem

$$\min_{\Delta u(\cdot)} \int_0^\infty \left(\beta^2 \Delta x(t)^T \Delta x(t) + \alpha^2 \Delta u(t)^2 \right) dt, \quad \alpha, \beta > 0, \quad (4a)$$

$$\text{s.t. } \Delta \dot{x}(t) = A \Delta x(t) + B \Delta u(t). \quad (4b)$$

Write the problem in standard LQR form by determining Q and R . State the optimal feedback law, and explain how it can be obtained.

- Calculate the optimal feedback gain K for $\alpha = 1$ and $\beta = 2$, then state the eigenvalues of the closed-loop system.

Exercise 5.3 Fishing problem

Let $x(t)$ be the fish population in a large lake at time t . Assuming that the harvesting rate is proportional to the current population, $x(t)$ can be approximated by the following differential equation

$$\dot{x}(t) = bx(t) \left(1 - \frac{x(t)}{c}\right) - u(t)x(t), \quad (5)$$

with the growth rate of the fish population $b > 0$, the carrying capacity of the lake $c > 0$, and the harvesting rate $u(t) \in [0, b]$. The objective is to maximize the revenue from fishing $pu(t)x(t)$, where $p > 0$ denotes the constant price per unit of fish.

- Compute and interpret the equilibria of the fish populations for $u = 0$ and $u = b$, respectively.
- Formulate a minimization problem for the objective from above. Then state the Hamiltonian, use Pontryagin's maximum principle to compute the optimal control law $u = \psi(\lambda)$ and interpret the adjoint variable.
- Under which condition does a singular case occur? Determine the singular control u_s and the corresponding fish population x_s .
- Interpret the singular solution with respect to the natural growth rate of the fish population.

Add-On 1 Maximum Principle for Time-Invariant Systems

Consider the following optimal control problem

$$\max_{u(\cdot)} x(t_f) + \int_0^{t_f} (1 - u(t))x(t) dt \quad (6a)$$

$$\text{s.t. } \dot{x}(t) = \alpha u(t)x(t), \quad x(0) = x_0 \quad (6b)$$

$$u(t) \in [0, 1], \quad \forall t \geq 0, \quad (6c)$$

with $0 < \alpha < 1$ and $x_0 > 0$. The state $x(t)$ can be interpreted as a resource, while $u(t)$ describes its growth fraction. The remaining fraction $1 - u(t)$ is harvested.

- Rewrite the problem as a minimization problem, state the Hamiltonian and derive the adjoint dynamics.
- Use Pontryagin's maximum principle to derive the optimal control law and define the switching function.
- Show that the switching can only occur once and determine the corresponding switching time t' .